DR. A.P.J. ABDUL KALAM TECHNICAL UNIVERSITY
LUCKNOW

Evaluation Scheme & Syllabus

FOR

B.Tech. 3rd Year

HANDLOOM & TEXTILE TECHNOLOGY

On

AICTE Model Curriculum

(Effective from the Session: 2022-23)

DR. A.P.J ABDUL KALAM TECHNICAL UNIVERSITY, LUCKNOW

STUDY AND EVALUATION SCHEME

SEMESTER- V													
Sl. No.	**Subject Codes**	**Subject**	**Periods**			**Evaluation Scheme**				**End Semester**		**Total**	**Credit**
			L	T	P	CT	TA	Total	PS	TE	PE		
1	**KHT-501**	Woven Fabric Manufacturing - I	3	1	0	30	20	44		100		150	4
2	**KHT-502**	Chemical Processing of Textiles - II	3	1	0	30	20	44		100		150	4
3	**KHT-503**	Quality Evaluation of Fibres and Yarns	3	1	0	30	20	44		100		150	4
4	**KHT-051/ KHT-052**	Garment Manufacturing Technology / Ethnic Handloom Fabrics	3	0	0	30	20	45		100		150	3
5	**KHT-053/ KHT-054**	Production of Manufactured Fiber / Engineering Mechanics for Textile Technologists	3	0	0	30	20	44		100		150	3
6	**KHT-551**	Chemical Processing of Textiles - II Lab	0	0	2				25		25	50	1
7	**KHT-552**	Fibre and Yarn Quality Evaluation Laboratory Lab	0	0	2				25		25	50	1
8	**KHT-553**	Garment Construction Lab	0	0	2				25		25	50	1
9	**KHT-554**	Mini Project or Internship Assessment*	0	0	2				50			50	1
10	**KNC501/ KNC502**	Constitution of India / Essence of Indian Traditional Knowledge	2	0	0	14	10	24		50			0
11		MOOCs (Essential for Hons. Degree)											
		Total										**950**	**22**

*The Mini Project or internship (4 weeks) conducted during summer break after IV semester and will be assessed during V semester.

SEMESTER- VI													
Sl. No.	Subject Codes	Subject	Periods			Evaluation Scheme				End Semester		Total	Credit
			L	T	P	CT	TA	Total	PS	TE	PE		
1	KHT-601	Woven Fabric Manufacturing - II	3	1	0	30	20	45		100		150	4
2	KHT-602	Fabric and Garment Quality Evaluation	3	1	0	30	20	44		100		150	4
3	KHT-603	Fabric Structure - II	3	1	0	30	20	44		100		150	4
4	KHT-061/ KHT-062	Knitting Technology / Home Textiles	3	0	0	30	20	44		100		150	3
5	KOE-060	Idea to Business Model	3	0	0	30	20	44		100		150	3
6	KHT-651	Textile Design and Color Lab	0	0	2				25		25	50	1
7	KHT-652	Fabric Structure Lab	0	0	2				25		25	50	1
8	KHT-653	Fabric Quality Evaluation Lab	0	0	2				25		25	50	1
9	KNC-601/ KNC-602	Essence of Indian Traditional Knowledge/Constitution of India	2	0	0	14	10	24		50			0
10		MOOCs (Essential for Hons. Degree)											
		Total										**900**	**21**

SEMESTER V

KHT 501 WOVEN FABRIC MANUFACTURING - I

L-T-P 3-1-0), total contact hours:- 60 Credits = 4

Course Objectives:

- To enable the students to understand about the weaving process of power loom, including different shedding mechanisms.
- To enable the students to understand detailed concepts related to Dobbies and auxiliary motion.
- To enable the students to understand detailed concepts related to Jacquard and its recent developments.
- To enable the students to understand detailed concepts related to gear, pulley & belt and loom shed calculations.
- To enable the students to understand detailed concepts related to cloth calculation.

UNIT 1

General description of plain power looms, introduction topower loom weaving process, primary, secondary and auxiliary motion of plain power looms, types of sheds, tappet shedding, over and under pick motions,Beating-up Mechanism – Eccentricity of sley, Timing and Synchronization of Primary motion,Temple and its utility, Negative and positive take up motion, five wheel and seven wheel take up motion, positive &negativelet-off motions.

Contact hours required = 9

UNIT 2

Scope & limitation of dobby, negative and positive dobby, right hand & left hand dobby & their pegging methods, Development in dobby, brief description of Cromption and Knowles dobby, cross border dobby, heald reversing motion. Warp protective mechanism- loose reed & fast reed, weft detection mechanisms-side and center weft fork motion.

Contact hours required = 9

UNIT 3

Types of jacquards and their principle of working, size and figuring capacity of jacquard, Single lift single cylinder Jacquard, Double lift single cylinder, Double lift double cylinder, split harness, cross border jacquards, Different system of harness tie- up, terry mechanism, Recent developments in jacquard weaving.

Contact hours required = 9

UNIT 4

Multiple box motion and their types, Pick-at will, pick and pick motion. Automatic loom: pirn change, shuttle change loom, detailed study of various motions of automatic looms, mechanical warp stop motions, electro-mechanical warp stop motion.

Contact hours required = 9

UNIT 5

Gear, pully & belt Calculations. Production and efficiency of machine e, take up and let off, Jacquard Design and Harness Calculations – Calculations related to Sett of harness, sett of warp, ends per repeat, size of repeat, number of repeats, symmetrical repeat setting for straight draft and pointed draft, number of cords per hook and casting out. Loom shed calculation – Efficiency, Production and Production Planning.

Contact hours required = 8

OUTCOMES:

Upon completion of the course the student will be able to:-

- Understand the working principle of power looms, dobby and jacquard.
- Understand the calculation related to machines, production and efficiency.

- Understand the calculation related to cloth (Various count system, cover).
- Understand the detailed concepts related to gear, pulley & belt and loom shed calculations.
- Understand detailed concepts related to Jacquard and its recent developments

REFERENCES:

1. Banerjee, P.K. (2014). Principles of Fabric Formation (1st Ed.). CRC Press. https://doi.org/10.1201/b17804
2. Marks, R., & Robinson, A. T. C. (2022). Principles of weaving. Manchester [England: Textile Institute.
3. Lord, P. R., & Mohamed, M. H. (1992). Weaving: Conversion of yarn to fabric.
4. Talukdar, M. K., Sriramulu, P. K., & Ajgaonkar, D. B. (1998). Weaving: Machines, mechanisms, management. Ahmedabad, India: Mahajan Publishers.
5. Omerod, A. (1983). Modern Preparation and Weaving Machinery. London: Butterworths.
6. Booth, J. E., & Textile Institute (Manchester, England). (1977). Textile mathematics: Vol. 3. Manchester: Textile Institute.
7. M. K. Talukdar. An Introduction to winding and Warping Testing Trade Press, Mumbai, 1982
8. Majumdar, A. B. H. I. J. I. T. (2020). Principles of Woven Fabric Manufacturing.

KHT 502 CHEMICAL PROCESSING OF TEXTILES – II

(L-T-P 3-1-0), total contact hours: - 44 Credits = 4

OBJECTIVE:

- To enable the students to understand the various aspects related to computer color matching.
- To enable the students to understand the theory of dyeing and printing and finishing of different natural and man-made fibers and their blends.
- To enable the students to understand the detailed concept related to dyeing terminology.
- To enable the students to understand basic concepts related to polyester, polyamide and acrylic dyeing.
- To give a broader concept of different terminologies of finishing like calendaring, shrinkage, improvement in serviceability etc.

UNIT 1

Theories of colour measurement, Beer–Lambert's law and Kubelka-Munk theory and their application in colour assessment and colour matching; whiteness and yellowness indices. Dyeing equilibrium; dye-fibre interaction; adsorption isotherm; dye affinity; heat of dyeing; half dyeing time Basic parameters of dyeing Viz. Percentage of shade, Percentage of exhaustion, percentage expression and effects of MLR.

Contact hours required = 9

UNIT 2

Basic characteristics of dyes and pigments; classification of dyes and principle of application of dyes; Principles and methods of application of Direct Dyes, reactive, vat, Solubilized vat, azoic with function of chemicals used and effect of process conditions on cellulosic fiber. Methods of dyeing of Wool and Silk with Acid, Chrome, Prematalized (1:1 & 1:2), and Reactive dyes with Process details such as time Temperature and pH, Effect of Electrolytes and use of leveling agents.

Contact hours required = 9

UNIT 3

Approaches for dyeing of polyester – Various methods of dyeing Polyester involving use of chemical and Thermal Energy (Carrier Dyeing and H.T.H.P. Dyeing, thermosol process) Dyeing of Polyamides with Disperse, Acid, Metal complex and Reactive dyes with details process parameters; Process parameter details of Dyeing Acrylic with Cationic and Disperse dyes. Dyeing of different blends.

Contact hours required = 9

UNIT 4

Methods and styles of printing; constituents of printing paste; printing with direct, reactive, vat, Solubilized vat, azoic, disperse, acid and basic dyes; printing with pigments, printing machines – roller printing, rotary printing and flat bed screen printing machines.

Contact hours required = 9

UNIT 5

Objects of Textile finishing, Factors affecting selection of Finishes, Classification of finishing processes, Mechanical Finishes Processes e.g. Calendering, and Compressive shrinkage process. Brief Outlines of Chemical finishing process for improvement in serviceability viz. Anti-crease

finishes, Softening and Stiffening, Flame Retardency, Water Proofing & Water Repellency Finishes, Introduction to Bio-finishing, Outlines of Harmful Chemicals in Wet Processing of Textiles, Introduction to the concept of Eco-friendly Wet processing. Continuous dyeing and beaching range, automation in process of dyeing, printing & finishing.

Contact hours required = 8

OUTCOMES:

Upon completion of the course the student will have knowledge on

- Understand various aspects of Colour, perception of colour and various terms related to dyeing.
- Understand the basics of different types of dyes and their application process.
- Understand the dyes for different fibers and their blend.
- Understand the methods and styles of printing and machines used for printing
- Understand the need of various finishes to the fabric and eco-friendly wet processing.

REFERENCES:

1. Shah H.S., and Gandhi R.S., "Instrumental Colour Measurement and Computer Aided Colour Matching for Textiles", Mahajan Book Publication, 1990, ISBN: 8185401004 / ISBN: 9788185401003
2. Choudhury A.K.R., "Modern concepts of colour and Appearance", Oxford and IBH publishing Ltd., 2000, ISBN: 1578080789 | ISBN-13: 9781578080786
3. Gulrajani M.L.(Ed.)., "Colour Measurement - Principles, advances and industrial applications", Wood head Publishing Ltd, 2010, ISBN: 1845695593 | ISBN-13: 9781845695590
4. Trotman E. R., "Dyeing and Chemical Technology of Textile Fibres", B.I Publishing Pvt.Ltd. New Delhi, 1994 , ISBN: 0471809101 | ISBN-13: 9780471809104
5. Shenai V.A., "Chemistry of Dyes and Principles of Dyeing", Sevak Publications, Mumbai, 1995.
6. Chakraborty J.N., "Fundamentals and practice in Colouration of Textiles", Wood head Publishing India Pvt Ltd, India, 2010, ISBN: 184569788X | ISBN-13: 9781845697884.
7. Mittal R.M., and Trivedi S.S., "Chemical Processing of Polyester/Cellulosic Blends", 2nd ed., Tata McGraw Hill, 2000.
8. Burkinshaw S. M., "Chemical Principles of Synthetic Fibre Dyeing", Springer-Science + Business Media, B.V., 2012, ISBN: 9401042632 | ISBN-13: 9789401042635.
9. Shenai V. A., "Technology of Printing", Sevak Publications, Mumbai, 1996.
10. Shahid, M., & Adivarekar, R. (2020). Advances in functional finishing of textiles. Singapore: Springer.
11. Veerapathran, R. (2014). Chemical processing of textile materials.

KHT 503 QUALITY EVALUATION OF FIBRES AND YARNS

(L-T-P 3-1-0), Total contact hours:- 44 Credits = 4

OBJECTIVE:

- To make the students understand the principle and method of working of equipments used for testing of fibres and yarns.
- To make the students understand the concepts related to tensile testing modes like CRT, CRE, CRL.
- To make the students understand the properties of different fibers& yarn technique.
- To make the students understand the concepts related to yarn evenness, twist and yarn strength.
- To make the students understand the concepts related to different aspects of yarn testing.

UNIT 1

Selection of samples for quality assessment – random and biased samples – squaring technique and zoning technique for fibre selection; yarn sampling - use of random numbers - sampling for various types of yarn tests**-** yarn strength, linear density, elongation, twist.

Contact hours required = 9

UNIT 2

Fibre testing, the fibre quality index and spinnablity; Fibre length and length uniformity measuring techniques. Tensile Strength Testing modes – CRT, CRE and CRL; Fibre strength and its relation to yarn strength. Fiber bundle strength tests.

Contact hours required = 8

UNIT 3

Fibre fineness – definition- comparison of various fibres – its importance in yarn manufacture; measurement techniques. Cotton fibre maturity, estimation by microscopic method - maturity ratio and index, estimation by other methods – optical, air flow differential dyeing; its importance in spinning. Fibre trash – influence on quality; measurement – principle and estimation of micro-dust for rotor spinning. High volume instrument for total fiber quality measurement.

Contact hours required = 9

UNIT 4

Yarn Appearance and grading, yarn mass evenness parameters – measurement – electronic mass evenness determination – Yarn fault classification –Yarn abrasion resistance – importance and measuring technique. Yarn hairiness – importance and assessment techniques. Twist in single and ply yarns –twist direction – twist factor – twist and yarn strength; twist measurement and breaking twist angle measurement. Analysis of data and test reports such as HVI, AFIS, Classimat.

Contact hours required = 9

UNIT 5

Tensile characteristics –study of strength, single yarn strength; Lea count strength product (CSP) and Corrected Count Strength Product (CCSP), elongation, work of rupture, initial modulus, work factor and yield point – determination of yield point. Stress-strain relations of natural and manmade fibres, influence of fibre structure, humidity and temperature on tensile characteristics. Time effects Study of creep phenomena. Elastic recovery and its relation to stress and strain of fibres; mechanical conditioning of fibres and its influence on elastic recovery. Load cycling and extension cycling-their effect on elastic recovery. Introduction about torsional and flexural rigidity of fibers.

Contact hours required = 9

OUTCOMES:

Upon completion of this course, the student shall be able to

- Understand the principle of operation of equipment's used for testing fibres and yarns
- Understand the Apply knowledge gained through this course, while operating the equipment's
- Understand the properties of different fibers and its role in spinning.
- Understand the concepts related to yarn evenness, twist and yarn strength.
- Understand the Analyze and interpret the results obtained from quality evaluating systems of fibre and yarns

REFERENCES:

1. Booth J.E., "Principle of Textile Testing", Butterworth Publications, London, 1989
2. Saville B.P., "Physical Testing of Textiles", Textile Institute, Manchester, 1998
3. Kothari V. K., "Testing and Quality Management", Progress in Textile Technology Vol.1, IAFL Publications, New Delhi, 1999
4. Arindam Basu, "Textile Testing", The South India Textile Research Association, Coimbatore, 2001
5. Vastra Reshe, Utpadan Upyog aur Visheshtayen D B Shakyawar, M K Singh, Abhishek Publications, Chandigarh India ISBN-10 : 8182475767
6. Characterization of Polymers and Fibres: M K Singh, A Singh Elsevier Publication Cambridge USA, 2021, eBook ISBN: 9780128242391
7. Slater K., "Physical Testing and Quality Control", The Textile Institute, Vol.23, No.1/2/3Manchester, 1993.
8. 1. Ruth Clock and Grace Kunz., "Apparel Manufacture- Sewn Product Analysis", Upper Sadle River Publications, New York, 2000
9. Pradip V. Mehta., "Managing Quality in the Apparel Industry", NIFT Publication, India, 1998

KHT 051 GARMENT MANUFACTURING TECHNOLOGY

(L-T-P 3-0-0), Total contact hours:- 45 Credits = 3

OBJECTIVES:

- To enable the students to understand the basics of garment manufacturing, pattern making & sewing and garment wet processing.
- To expose the students to various apparel production systems, material handling and processing of garment.
- To provide knowledge of structure of garment industry.
- To deliver the details of technical requirements of garment manufacturing process
- To be able to select fabrics by appropriate testing for making garments.

UNIT 1

Anthropometry, mass-production, mass-customization; pattern making, grading, marker planning, spreading & cutting. Different types of seams and stitches; single needle lock stitch machine - mechanism and accessories; needle functions, special needles, needle size, numbering, needlepoint; sewing thread construction, material, thread size, packages.

Contact hours required = 9

UNIT 2

Outline of various Apparel Production Systems and Material Handling. Quality control systems in garment manufacturing. Garment dyeing, printing and finishing; pressing categories and equipment, packing.

Contact hours required = 9

UNIT 3

Raw material- in process and final inspection; needle cutting; sewability of fabrics; strength properties of apparel; dimensional changes in apparel due to laundering, dry-cleaning, steaming and pressing; care labeling of apparel.

Contact hours required = 9

UNIT 4

Labels, linings, interlinings, wadding, lace, braid, elastic, hook and loop fastening, shoulder pads, eyelets and laces, zip fasteners, buttons. Pattern alteration technique, principle of fittings, selection of fabrics, Trims and Accessories, Methods of fabric inspection.

Contact hours required = 9

UNIT 5

Various production system with their characteristics, lay out planning, Work study methods, motion and time study, computer Integrated production planning and management systems. Concepts of CAD, computer aided embroidery designs, computer integrated manufacturing (CIM) in the garment manufacturing and technological advancement in garment manufacturing.

Contact hours required = 9

OUTCOMES:

Upon completion of the course, the students will

- Understand the know about pattern making, market planning, cutting and sewing of apparels of garments.
- Understand the basics of garment manufacturing, garment wet processing.
- Understand the different types of seams and stitches; needle; sewing thread construction, packages.
- Understand the labels, linings, interlinings, wadding, lace, braid, elastic, hook and loop fastening, shoulder pads, eyelets and laces, zip fasteners, buttons.
- To enable the students to understand garment dyeing, printing and finishing; pressing categories and equipment, packing.

REFERENCES:

1. Carr H., and Latham B., "The Technology of Clothing Manufacture", Blackwell Science Ltd., Oxford, 1994.
2. Gerry Cooklin, "Introduction to Clothing Manufacture" Blackwell Science Ltd., 1995.
3. Harrison.P.W Garment Dyeing, The Textile Institute Publication, Textile Progress, Vol .19 No.2,1988.
4. Winifred Aldrich., "Metric Pattern Cutting", Blackwell Science Ltd., Oxford, 1994
5. Peggal H., "The Complete Dress Maker", Marshall Caverdish, London, 1985
6. Jai Prakash and Gaur R.K., "Sewing Thread", NITRA, 1994
7. Ruth Glock, Grace I. Kunz, "Apparel Manufacturing", Dorling Kindersley Publishing Inc., New Jersey, 1995.
8. PradipV.Mehta, "An Introduction to Quality Control for the Apparel Industry", J.S.N. Internationals, 1992.
9. In Nayak, R., In Padhye, R., & Textile Institute (Manchester, England),. (2015). Garment manufacturing technology.

KHT 052 ETHNIC HANDLOOM FABRICS

(L-T-P 3-0-0), Total contact hours:- 44 Credits = 3

OBJECTIVES:

- To enable the students to know the importance of textures and special features of ethnic handloom fabrics.
- To enable the students to know the richness and importance of traditional handloom fabrics.
- To enable the students to know the specialty of Salem silk, Angavastram, Handloom Lungies, and stoles.
- To enable the students to know the details of different shawls, carpets, durries and furnishings made up of Kannur.
- To enable the students to know the details of handloom articles reserved for exclusive production on handlooms.

UNIT 1

Study of construction details , loom particulars, production techniques (raw material to end product) , and specialty of Kanchipuram Saree, Chanderi Saree, Balaramapuram Saree , Venkatagiri Saree, Banaras Brocade and Saree, Paithani Saree, Baluchari Saree, maheshwari saree, Jamdani, AwadhJamdani, Muga silk saree of Assam.

Contact hours required = 9

UNIT 2

Study of construction details, loom particulars, production techniques (raw materials to end product) and Specialty of Patola Saree of Gujarat, Ikats of Andhra Pradesh and Orissa, Kalam Kari , Tie and Dye fabrics of Rajasthan-Bhandhej, Mothda&Laheriya.

Contact hours required = 8

UNIT 3

Construction details, loom particulars, production techniques(Raw Marterial to end products) and specialty of Salem silk and cotton Dhotis, silk and Cotton Angavastram, Handloom Lungies, Study of stoles- Kinnori stole, Pashmina stole, Tangalia stole, Bagru block print stoles.

Contact hours required = 9

UNIT 4

Study of shawl- Himroo Shawl, Kullu Shawl , Naga Shawl , Wool Jamawar Shawls, Cashmere Check Shawl, Pashmina Shawls, Carpets – Hand Woven Pile Carpets, Plain & Figured Durries , PanipetKhes and Tapestry – Furnishing & made –ups of Kannur.

Contact hours required = 8

UNIT 5

Study of handloom articles reserved for exclusive production on handlooms Saree, Dhothi, Towel and Gamcha, Angavastarm,Lungi, Khes, Bed sheets, Bedcover, Counterpane Furnishing (including Tapestry, Upholstery) Jamakkalam, Durry or Durret, Dress material, Barrack Blankets and Kambalor Kamblies, Shawl, Loi, Muffler, Pankhi, Woollen Tweet, Chaddar , Mekhala or Phanek.

Contact hours required = 10

OUTCOMES:

Upon completion of this course, the student will be able to:

- Understand the of textures and special features of ethnic handloom fabrics.
- Understand the richness and importance of traditional handloom fabrics.
- Understand the to know the specialty of Salem silk, Angavastram, Handloom Lungies, and

stoles.

- Understand the details of different shawls, carpets, durries and furnishings made up of Kannur.
- Understand the details of handloom articles reserved for exclusive production on handlooms.

REFERENCES:

1. S.Rangarajan, "HEPC product manual", Handloom Export Promotion council, Chennai, 1994
2. R.T.A KapurChishti&Ambasanyal, "Saris of India", Wiley Eastern Ltd, New Delhi, 1989
3. Dr.Moti Chandra, "Ikat sarees of Orissa and Andhrapradesh", Calico museum of Textiles, Ahmedabad,1974
4. "India Handlooms – Tantavi Tradition & Trend catalogue" NCTD, O/o. Development Commissioner for Handlooms, New Delhi, 2015
5. "Indian Handlooms – Tantavi heritage catalogue" NCTD, O/o. Development Commissioner for Handlooms, New Delhi, 2012-13
6. B. K. Behera, Traditional Textile Designs – Gold line Publisher - New Delhi , 200

KHT 053 PRODUCTION OF MANUFACTURED FIBRE

(L-T-P 3-0-0), Total contact hours:-44 Credits =3

OBJECTIVES:

- To enable the students to learn about the polymer rheology and the laws, and various spinning techniques of polymers.
- To expose the students to advances in the melt and dry spinning process.
- To expose the students to advances in the solution spinning process.
- To brief students about drawing, heat setting and texturizing.
- To enable the students to learn about the advanced spinning, specialty fibers and its applications.

UNIT 1

Transport Phenomena in Fibre Manufacturing- Heat and mass; Polymer rheology-Newtonian and non-newtonian fluids, factors affecting shear viscosity; conditions of fibre forming polymers; Melt instabilities.

Contact hours required = 9

UNIT 2

Melt Spinning- Polymer Selection and Preparation, equipment, properties and fibre manufacturing of polyester, polyamide and polypropylene fibers, Dry spinning technique.

Contact hours required = 8

UNIT 3

Classification of solution spinning, Polymer Selection and Preparation, equipment, properties and fibre manufacturing of acrylic, polyurethane and regenerated cellulose fibres.

Contact hours required = 9

UNIT 4

Neck drawing, drawing systems, influence of drawing on structure and properties of fibres; Types of heat setting, influencing parameters on heat setting, influence of heat setting on fibre behavior,Important of Spin finish composition&application, classification& different techniques of texturizing.

Contact hours required = 9

UNIT 5

Liquid crystal spinning; Gel spinning; Profile fibres, hollow & porous fibres; Speciality fibres polyglycolic acid, polylactic acid, chitosan fibres preparation properties and applications.
Introduction and use of high performance fibers- Aramids, Carbon, Glass etc.

Contact hours required = 9

OUTCOMES:

Upon completion of this course, the student shall be able to understand

- Understand the polymer rheology and the laws, and various spinning techniques of polymers.
- Understand the in the melt and dry spinning process.
- Understand the advances in the solution spinning process.
- Understand the drawing, heat setting and texturizing.
- Understand the Various spinning techniques of polymers parameter involved in spinning synthetic yarn

REFERENCES:

1. Kothari V. K., "Textile Fibres: Development and Innovations", Vol. 2, Progress in Textiles, IAFL Publications, New Delhi, 2000
2. Characterization of Polymers and Fibres: M K Singh, A Singh Elsevier Publication Cambridge USA, 2021, eBook ISBN: 9780128242391
3. Nakasjima (English edition, edited by Kajiwara K. and McIntyre J. E.), "Advanced Fibre Spinning Technology", Wood head Publication Ltd., England, 1994.
4. Gupta V. B. and Kothari V. K. (Editors), "Manufactured Fibre Technology", Kluwer Academic Publishers, 1997.
5. Cook J. G., "Handbook of Textile Fibres: Vol. 2: Man Made Fibres", The Textile Inst., 5th Ed. 1984.
6. Srinivasa Murthy H. V., "Introduction to Textile Fibres", Textile Association, India, 1987.
7. Uddin, F. , (Ed.). (2019). Textile Manufacturing Processes.
8. Gupta, V. B., & Kothari, V. K. (2012). Manufactured Fibre Technology. Dordrecht: Springer Netherlands.

KHT 054 ENGINEERING MECHANICS FOR TEXTILE TECHNOLOGISTS

(L-T-P 3-0-0), Total contact hours:-44 Credits =3

OBJECTIVE:

- To develop capacity to predict the effect of force and motion in the course of carrying out the design functions of engineering.
- To enable students to study free body diagram, equilibrium of Rigid bodies in three dimensions.
- To enable students to study Centroids, theorems of pappus, area moments of inertia.
- To enable students to study curvilinear motion-Newton's laws of motion, Impact of elastic bodies.
- To enable students to study Friction force, rolling resistance, General Plane motion of simple rigid bodies such as cylinder, disc/wheel and sphere.

UNIT 1

Introduction – Units and Dimensions – Laws of Mechanics – Lami's theorem, Parallelogram and triangular Law of forces — Vectorial representation of forces – Vector operations of forces - additions, subtraction, dot product, cross product – Coplanar Forces – rectangular components – Equilibrium of a particle – Forces in space – Equilibrium of a particle in space – Equivalent systems of forces – Principle of transmissibility .

Contact hours required = 9

UNIT 2

Free body diagram – Types of supports –Action and reaction forces –stable equilibrium – Moments and Couples – Moment of a force about a point and about an axis – Vectorial representation of moments and couples – Scalar components of a moment – Varignon's theorem

Single equivalent force -Equilibrium of Rigid bodies in two dimensions – Equilibrium of Rigid bodies in three dimensions

Contact hours required = 8

UNIT 3

Centroids and centre of mass– Centroids of lines and areas - Rectangular, circular, triangular areas by integration – T section, I section, - Angle section, Hollow section by using standard formula – Theorems of Pappus -Area moments of inertia of plane areas – Rectangular, circular, triangular areas by integration – T section, I section, Angle section, Hollow section by using standard formula – Parallel axis theorem and perpendicular axis theorem –Principal moments of inertia of plane areas – Principal axes of inertia-Mass moment of inertia –mass moment of inertia for prismatic, cylindrical and spherical solids from first principle – Relation to area moments of inertia.

Contact hours required = 9

UNIT 4

Displacements, Velocity and acceleration, their relationship – Relative motion – Curvilinear motion -Newton's laws of motion – Work Energy Equation– Impulse and Momentum – Impact of elastic bodies.

Contact hours required = 9

UNIT 5

Friction force – Laws of sliding friction – equilibrium analysis of simple systems with sliding friction–wedge friction-. Rolling resistance -Translation and Rotation of Rigid Bodies – Velocity and acceleration – General Plane motion of simple rigid bodies such as cylinder, disc/wheel and sphere.

Contact hours required = 9

OUTCOMES:

Upon completion of this course, the student shall be able to understand

- The Ability to explain the differential principle applies to solve engineering problems dealing with force, displacement, velocity and acceleration.
- The Ability to analyses the forces in any structures.
- The Ability to solve rigid body subjected to dynamic forces.
- The develop capacity to predict the effect of force and motion in the course of carrying out the design functions of engineering.
- The enable students to study free body diagram, equilibrium of rigid bodies in three dimensions.

REFERENCES:

1. Beer, F.P and Johnson Jr. E.R. "Vector Mechanics for Engineers (In SI Units): Statics and Dynamics", 8th Edition, Tata McGraw-Hill Publishing company, New Delhi (2004)
2. Vela Murali, "Engineering Mechanics", Oxford University Press (2010)
3. Hibbeller, R.C and Ashok Gupta, "Engineering Mechanics: Statics and Dynamics", 11th Edition, Pearson Education (2010).
4. Irving H. Shames and Krishna MohanaRao. G., "Engineering Mechanics – Statics and Dynamics", 4th Edition, Pearson Education (2006).
5. J. L. Meriam and L.G.Kraige, "Engineering Mechanics - Statics - Volume 1, Dynamics Volume 2,Third Edition, John Wiley & Sons, (1993).

KHT 551 CHEMICAL PROCESSING OF TEXTILE - II LABORATORY

(L-T-P 0-0-2),Credit =1

LIST OF EXPERIMENTS

1) Dying of cotton/ cellulosic fibers with Direct, Reactive, Vat, Sulphur, Azoic, and Solubilizedvat dyes.

2) Dying of wool and silk fibers with Acid, Chrome, Prematalized(**1:1 & 1:2**) And Reactive Dyes.

3) Dying of Polyester, Acrylic and Nylon fibers with Acid, Disperse, Basic, Acid, Metal Complex and Reactive Dyes.

4) Printing of cotton fabrics with Pigments by Direct style using Blocks / screens

5) Printing of silk fabrics with Acid Dyes by Direct style using Blocks / Screens

6) Printing of polyester fabrics with Disperse dyes by Direct style using Blocks / Screens.

7) Discharge style printing of cotton fabrics dyed with Reactive dyes

8) Discharge style printing of silk fabrics dyed with Acid/Metal complex dyes

9) Printing of cotton fabric by resist style (Batik printing) using Reactive cold brand / cold dyeing Vat dyes.

10) Tie and dye (Resist style) of cotton yarn and fabric using Reactive / Vat dyes

11) Printing of Polyester/Cotton blends with Disperse/Reactive dyes by direct style using Block / Screen.

12) Crease resistance finishing on cotton fabric

13) Water resistance finishing on cotton fabric

14) Determination of Dimensional Change of cotton fabrics

KHT 552 FIBRE AND YARN QUALITY EVALUATION LABORATORY

(L-T-P 0-0-2),Credit =1

LIST OF EXPERIMENTS

1) To Measure fibre fineness by air flow method.

2) To find out fibre length by Baer Sorter & analytical digital fibrograph.

3) To find out Fiber maturity test by Caustic Soda Swelling method.

4) To studies fibre trash by Shirley trash analyzer.

5) To find out fibre bundle strength by using Pressley fibre bundle strength tester.

6) To find out linear density of sliver, Roving and yarn.

7) To find out the yarn strength by using Single yarn strength tester.

8) Fine out yarn lea strength of yarn by using Lea strength tester.

9) To find the Single and folded yarn twist by using yarn twist tester.

10) Measurement of yarn Unevenness by using USTER evenness tester.

11) To fine out Yarn appearance by using Yarn Appearance Board Winder

KHT 553 GARMENT CONSTRUCTION LABORATORY

(L-T-P 0-0-2),Credit =1

LIST OF EXPERIMENTS

1) To study the classes of Stitches &its properties.

2) Study of seam classification and common uses.

3) To create various type of Stitches on the fabric surface by using different type of yarn.

4) Formation of superimposed seam, lapped seam, bound seam and flat seam in the fabric.

5) To perform the Button hole making and button stitching.

6) Passes of material though Feed-of-the-arm machine with define different machine part.

7) To create design & draft for different type of paper Pattern for Men's wear.

8) To perform the fabric Cutting method for Men's wear.

9) To perform method of Sewing and finishing Men's wear.

10) To create design & draft for different type of paper Pattern for Women's wear.

11) To perform the fabric Cutting method for Women's wear.

12) To perform method of Sewing and finishing of women's wear.

SEMESTER VI

KHT 601 WOVEN FABRIC MANUFACTURING – II

(L-T-P 3-1-0), Total contact hours:-45 Credits =4

OBJECTIVE:

- To enable the students to understand the production techniques of terry weaving, and fabric formation by non-conventional weaving machines.
- To know about the details related to Projectile weaving machine and Rapier weaving machine.
- To know about the details related to Air jet weaving machine, Water-jet weaving machine and Multi phase weaving machine.
- To know about the details related to Cloth cover- Derivation and calculation.
- To know about the Textile calculation, related to fabric construction.

UNIT 1

Introduction to terry weaving, Classification of terry fabrics, different mechanism for terry manufacturing, Properties and performance of terry fabric.Projectile Weaving Machine:Basic principle of projectile loom, sequence of weft insertion, camdrive shedding mechanism, beat-up, torsion bar picking, loom timing, checking of gripper, let-off and take-up motion, selvedge formation, Energy equation of torsion bar, velocity and acceleration of the projectile.

Contact hours required = 9

UNIT 2

Rapier Weaving Machine:Method of weft insertion system- Gabler, dewas types of rapier, rigid, flexible, telescope, weft insertion sequencein rapier weaving process. Different methods to drive the rapier head. Single phase double acting rapier.Velocity of the rapier. Loom timing.

Contact hours required = 9

UNIT 3

Air jet weaving machine: Problem in air jet weft insertion, Sequence of weft insertion in air jetloom. Design of confuser guide, design of profile reed and relay jet. Loom timing.Water-jet Weaving machine: Weft insertion mechanism, quality of warp required for water jet,selvedge formation, quality of water, problem of water-jet loom.Multi-phase Weaving Machine: Basic concept of multiphase weaving.

Contact hours required = 9

UNIT 4

Diameter of yarns Ashenhurst's& Pierce's formula for estimation of diameter of yarns. Cloth cover- Derivation and calculation. Fractional Cover, Percentage cover and Cover factor – Calculation for light medium and heavy fabrics.

Contact hours required = 9

UNIT 5

Calculation related to maximum setting of cloth. Determination of Ends per inch and picks per inch while changing count or weave and its combination to maintain the same level of compactness.Costing of Fabrics – Handloom-made, Power-loom-made and Mill-made – involving cotton, polyester, silk, wool, etc.

Contact hours required = 9

OUTCOMES:

Upon completion of the course the student will be able to:-

- Understand production techniques of terry weaving.
- Understand to Fabric formation by non-conventional weaving machines like, projectile, rapier, water-jet and multi-phase weaving machines.
- Understand to Textile calculation, related to fabric construction.
- Understand to enable the students to understand the production techniques of terry weaving, and fabric formation by non-conventional weaving machines.

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1. Grosicki Z. J. "Watson's Textile Design and Colour: Vo.1, Woodhead Publications, Cambride England, 2004.
2. Grosicki Z. J., "Watson's Advanced Textile Design and Colour", Vol.II, Butterworths, London, 1989.
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4. Lord P R and Mohamad M H, "Weaving: Conversion of Yarn to Fabric", Merrow Technical Library, UK, 1988.
5. Ormerod A, "Modern preparations and weaving machinery", Buttersworth and co., UK, 1983.
6. Talavasek O and Svaty V, "Shuttleless weaving machine", Elsevier Scientific Publishing Co., Amsterdam, 1981.
7. Lunenschloss J and Albrecht W, "Non-Woven Bonded Fabric", Ellis and Horwood Ltd., UK, 1985
8. Majumdar, A. (2016). Principles of Woven Fabric Manufacturing (1st ed.).
9. Thangamani, K., &Sundaresan, S. (2022). Fabric manufacturing technology: Weaving and knitting.

KHT602 FABRIC AND GARMENT QUALITY EVALUATION

(L-T-P 3-1-0), Total contact hours:-44 Credits =4

OBJECTIVE:

- To enable the students to learn about the construction of fabrics and evaluation of fabric properties and their importance.
- To enable the students to learn about the Tensile strength measurement, Tear strength, Bursting strength and its measurement.
- To enable the students to learn about the Fabric stiffness, Drape ability, Crease recovery, Wrinkle recovery, Fabric abrasion resistance, Fabric pilling resistance.
- To enable the students to learn about hysteresis; compression and decompression; surface roughness and friction measurements; Textile product leveling; Uster standards, AITCC, JIS and ASTM.
- To enable the students to learn about Fabric inspection, classification of fabric defects, MIL standards and final inspection, Colour matching, Analysis of KES and FAST data.

UNIT 1

Basic fabric particulars – Measurement of ends and picks per inch, count of warp and weft, determination of the type of weave, measurement of length, width, thickness and Area density (GSM); warp and weft crimp measurements for spun and filament yarn fabrics, the cover factor calculations; Fabric sampling techniques.

Contact hours required = 9

UNIT 2

Tensile strength measurement – raveled strip test and grab test – mechanical and electronic measuring systems. Tear strength – importance – measuring systems. Bursting strength and its measurement. Ballistic impact strength. Universal tensile tester - principle and operation

Contact hours required = 8

UNIT 3

Fabric stiffness – principle of measurement of flexural rigidity; Drapability – measurement of drape coefficient; Crease recovery measurement techniques. Wrinkle recovery assessment using standard grades; Principle and functioning of air permeability testers, water repellency, contact angle and fabric shrinkage testing; Fabric abrasion resistance – measuring technique; Fabric pilling resistance – methods of determination.

Contact hours required = 9

UNIT 4

Fabric bending hysteresis testing; Shear hysteresis measurements; Fabric compression and decompression behavior; Fabric surface roughness and friction measurements; Fabric tensile hysteresis measurements; Fabric flame resistance testing methods; Moisture and thermal characteristics. Textile product leveling. International quality parameters and standards like Uster standards, AITCC, JIS and ASTM.

Contact hours required = 9

UNIT 5

Fabric inspection – Manual, semi-automatic and Automatic Inspection systems, classification of fabric defects, independent product quality certification, acceptable quality level, MIL standards and final inspection. Quality assessment of garments - cutting, sewing, pressing, finishing and packaging defects. Assessment of comfort, Colour matching, Analysis of KES and FAST data.

Contact hours required = 9

OUTCOMES:

Upon completion of this course, students would be able to

- Understand Evaluate the constructional parameters of fabrics
- Understand the principle of measurement of fabric characteristics
- Understand Analyze the various reports generated during quality evaluation of fabric
- Understand to enable the students to learn about the Tensile strength measurement, Tear strength, Bursting strength and its measurement.
- Understand to enable the students to learn about the Fabric stiffness, Drape ability, Crease recovery, Wrinkle recovery,

REFERENCES:

1. Booth J.E., "Principle of Textile Testing", Butterworth Publications, London, 1989
2. Saville B.P., "Physical Testing of Textiles", Textile Institute, Manchester, 1998
3. Kothari V. K., "Testing and Quality Management", Progress in Textile Technology Vol.1, IAFL Publications, New Delhi, 1999
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KHT 603 FABRIC STRUCTURE II

(L-T-P 3-1-0), Total contact hours:-44 Credits =4

OBJECTIVE:

- To enable the students to understand the construction of double layer and pile fabrics and its production techniques.
- To enable the students to understand the double cloth, stripes, warp and weft patterning; treble width plain cloth – interlacement diagram and its graphical representation; Warp and weft backed cloths.
- To enable the students to understand the Pile fabrics, Velvet; two beams and 'W' binding.
- To enable the students to understand the Velveteen, Corduroys, Distorted thread structures, Leno and gauze fabrics, Bottom douping and Top douping, Easer and shaker device; Russian Cord.
- To enable the students to understand the Damask, Figured double cloth structure, extra warp and extra weft, Tapestry.

UNIT1

Double cloth classification – Step by step construction of self-stitched double cloth, reversible and non-reversible varieties using twill and sateen; Centre stitched double cloth; double width plain cloth – interlacement diagram and its graphical representation; plain Tubular cloth – interlacement diagram and its graphical representation; design, draft, denting, peg/tie-up plan and thread interlacing diagram of above weaves.

Contact hours required = 9

UNIT2

Thread interchanging double cloth, warp thread interchanging double cloth and weft thread interchanging double cloth, Cloth interchanging double cloth using plain and twill, creation of column stripes, row stripes and check effects using warp and weft patterning; wadded double cloth–warpwaddingandweftwadding;Treblewidthplaincloth–interlacementdiagramanditsgraphical representation; Treble cloth using twill, satin, and sateen; Backed cloths – Salient features, Warp and weft backed cloths - Reversible and non-reversible warp backed cloth using twill and sateen; design, draft, denting, peg/tie-up plan and thread interlacing diagram of above weaves.

Contact hours required = 9

UNIT 3

Pile fabrics – Salient features, classification of pile fabrics, Terry piles – salient features, terry mechanism; classification of terry pile structures – 3 pick, 4 pick, 5 pick and 6 pick terry, graphical representation and thread interlacement diagram; terry pile on one side of fabric and on both sides, stripe and check effects – terry pile on one side and both sides; design, draft, denting, peg plan, tie-up and thread interlacing diagram of aboveweaves.Velvet - Warp pile fabrics produced with the aid of wires, loop piles and cut piles production techniques – simultaneous insertion of pick and wire and the twin shed technique; techniques of anchoring of piles using two beams and by 'W' binding; design,

draft, denting, peg plan, tie-up and thread interlacing diagram of above weaves.

Contact hours required = 9

UNIT 4

Velveteen- salient features, manufacturing technique and processes involved; All over velveteen – Plain back and Twill back velveteen, Corduroys; design, draft, denting, peg/tie-up plan and thread interlacing diagram of above weaves. Distorted thread structures, Leno and gauze fabrics – Salient features, open, crossed and plain sheds in leno weaving; Bottom douping and Top douping, Easer and shaker device; thread interchanging diagram of leno structures, Combination of Leno, plain and twill weaves – Russian Cord – Net leno -drafting and lifting plans.

Contact hours required = 9

UNIT 5

Figured single cloth designing and enlargement, Damask- silent feature, designing and simplified enlargement technique, Figured pique- structure, 2pick, 3pick, 4pick, Introduction of wadding weft in 3 pick and 4 pick piques. Figured double cloth structure, Figured terry piles- 3 pick & 4 Pick, Figured warp backed cloth – Figured weft backed cloth. Figuring with extra threads – extra warp and extra weft, Tapestry- 3 pick & 4 pick reversible and non-reversible.

Contact hours required = 9

OUTCOMES:

Upon completion of the course the student will be able to:-

- Understand the double cloth, stripes, warp and weft patterning; treble width plain cloth – interlacement diagram and its graphical representation; Warp and weft backed cloths.
- Understand the Pile fabrics, Velvet; two beams and 'W' binding.
- Understand the Velveteen, Corduroys, Distorted thread structures, Leno and gauze fabrics
- Understand the Damask, Figured double cloth structure, extra warp and extra weft, Tapestry.

REFERENCES:

1. Grosicki Z. J. "Watson's Textile Design and Colour: Vo.1, Woodhead Publications, Cambride England, 2004.
2. Grosicki Z. J., "Watson's Advanced Textile Design and Colour", Vol.II, Butterworths, London, 1989.
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8. Gokarneshan, N. (2004). Fabric structure and design. New Delhi, India: New Age International (P) Ltd., Publishers.
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10. Read, J. (2011). Elementary textile design and fabric structure.

KHT061 KNITTING TECHNOLOGY

(L-T-P 3-0-0), Total contact hours:-44 Credits =3

OBJECTIVES:

To make the students to understand:-

- Knitting, knitting industry. Comparison of fabric properties, classification of knitting
- Fundamentals of knitting
- Types of knitting processes in detail
- Functioning of components of knitting machine
- Warp knitting-basic principles and Uses of warp knitted fabrics in technical applications.

UNIT 1

Introduction to Knitting-Reasons for the growth of the knitting industry. Comparison of fabric properties - wovens, knits and bonded fabrics; classification of knitting processes – weft knit & warp knit; yarn quality requirements for knitting. Preparation of staple yarns for weft and warp knitting.

Contact hours required = 9

UNIT 2

Fundamentals of Knitting-General definitions and principles of knitting; Types of knitting needles – Bearded, Latch & Compound Needle. Elements of knitted loop structure.

Contact hours required = 9

UNIT 3

Weft Knit Structures-Basic weft knitted structures and their production - plain, rib, interlock and purl; Fundamentals of formation of knit, tuck and float stitches; factors affecting the formation of loop; effect of loop length and shape on fabric properties; Analysis of various types of weft knitted structure. Weft knitted fabric geometry.

Contact hours required = 9

UNIT 4

Weft Knitting Machines-Construction, Characteristics and working of circular knitting machines used for the production of basic structures; production of derivatives of weft knitted structures; needle control in circular knitting machines; quality control in knitted fabric production; production calculation. Basic principles and elements of flat knitting machines; different types of flat knitting machines - manual, mechanical and computer controlled; production of various weft knitted structures using flat knitting machines.

Contact hours required = 9

UNIT 5

Warp Knitting-Basic principles; elements of warp knitted loop – open loop, closed loop; warp knitting elements chain link, chain links for simple patterns, guide bar movement mechanism,. Tricot and Rachel warp knitting machines. Principles of double needle bar patterning, Terry pile fabric production. Let off system; run in value based on the lapping diagram; take up system; theoretical concepts of warp knitted loop configuration.; Uses of

warp knitted fabrics in technical applications.

Contact hours required = 9

OUTCOMES:

Upon completion of this course, the student shall be able to understand the

- Understand the Principle of knitting by different types of knitting machines
- Understand the Structure and properties of fabric produced by different knitting machines
- Understand the Fundamentals of knitting
- Understand the Types of knitting processes in detail
- Warp knitting-basic principles and Uses of warp knitted fabrics in technical applications

REFERENCES:

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2. Ajgaonkar D.B., "Knitting Technology", Universal Publishing Corporation, Mumbai, 1998, ISBN: 81-85027-34-X
3. Gajjap B.J., "Handbook of warp Knitting Technology", Textile Institute, Manchester, 2004, ISBN: 1 85573 7701
4. Chandrasekhar Iyer, Bernd Mammel and Wolfgang Schach., "CircularKnitting", Meisenbach GmbH, Bamberg, 1995, ISBN: 3-87525-066-4
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9. Paling D.F., "Warp Knitting Technology", Columbine Press, U.K, 1966
10. Charles Reichman, "Wool and Synthetic Knitwear Handbook", National Knitted Outerwear Association, U.S.A, 1967
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KHT062 HOME TEXTILES

(L-T-P 3-0-0), Total contact hours:-44 , Credits =3

OBJECTIVES:

- To enable the students to learn about the recent developments in furnishing, factors affecting selection of home Furnishings.
- To enable the students to learn about the recent developments in floor covering and other home textile products various kinds of materials used in home textile.
- To enable the students to learn about the recent developments in curtains and draperies.
- To enable the students to learn about the recent developments in use of Colors, design & texture in home furnishing, Wall Hangings, Cushion, Cushion Covers, Bolster and Bolster Cover.
- To enable the students to learn about the recent developments in Bed Linens, Sheets, Blankets, Blanket Covers, Comforts, Comfort Covers, Bed Spreads, Mattress and Mattress Covers, Pads,

UNIT 1

Furnishings-Developments in Textile Furnishing; Type of Furnishings Materials – Woven and non-woven; Factors affecting selection of Home Furnishings.

Contact hours required = 8

UNIT 2

Floor Coverings-Recent Developments in manufacturing of floor coverings -Hard Floor Coverings, Resilient Floor Coverings, Soft Floor Coverings, Carpet,Rugs, Cushion and Pads; Care of floor coverings.

UNIT 3

Curtains and Draperies-Advances in Home decoration - Draperies – Choice of Fabrics, Curtains – Types of Developments in Finishing of Draperies; Developments in tucks and Pleats; uses of Drapery Rods, Hooks, Tape Rings and Pins.

Contact hours required = 9

UNIT 4

Home Furnishing-Advances in period style in, Different styles, and use of Colours, design & texture in home furnishing. Developments in living room furnishing including upholstery, Wall Hangings, Cushion, Cushion Covers, Bolster and Bolster Cover.

Contact hours required = 9

UNIT 5

Bed Linens-Advances in the production of - Different Types of Bed Linen**,** Bed skirts,Pillow cover,Throw, Sheets, Blankets, Blanket Covers, Comforts, Comfort Covers, Bed Spreads, Mattress and Mattress Covers, Pads, Pillows.

Contact hours required = 9

OUTCOMES:

Upon completion of this course, the student shall be able to:-

- Understand the production method of different types of home textile products.
- Understand to the recent developments in furnishing, factors affecting selection of home Furnishings.
- Understand to the recent developments in floor covering and other home textile products various kinds of materials used in home textile.
- Understand to enable the students to learn about the recent developments in curtains and draperies.
- Understand to enable the students to learn about the recent developments in use of Colors, design & texture in home furnishing,

REFERENCES:

1. Alexander.N.G., "Designing Interior Environment", Mas Court Brace Covanorich, Newyork, 1972
2. Donserkery.K.G., "Interior Decoration in India", D. B. Taraporeval Sons and Co. Pvt. Ltd., 1973
3. Wingate I.B. &Mohler J.F., "Textile Farbics& Their Selection", Prentice Hall Inc., New York, 1984.
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OPEN ELECTIVE-I

KOE-060 IDEA TO BUSINESS MODEL

(L-T-P 3-0-0), Total contact hours:-44 Credits =3

Course Objectives:

1. This course can motivate students to have an overall idea how to start and sustain a business enterprise.
2. The students will earn basics of choosing an idea of a business model.
3. The core areas of choosing a business model are encompassed with Entrepreneurship development, PPC & communication system. The students will thus develop basic competencies how to run a business enterprise.

UNIT 1

Introduction- Search for a business idea- How to choose an idea- Product idea- selection of product- The adoption process- Product innovation- Production , planning and development strategy- New product idea.

Contact hours required 8

UNIT 2

Introduction to Entrepreneurship - Meaning and concept of entrepreneurship- Difference between Entrepreneurship & wage employment - Functions of an Entrepreneur.- Entrepreneur vs Manager- role of entrepreneurship in economic development – Barriers to entrepreneurship.

Contact hours required 9

UNIT 3

The Entrepreneur - types of entrepreneurs- Competencies required to become an entrepreneur - Creative and Design Thinking, the entrepreneurial decision process- The process of Entrepreneurial development program (EDP)- Evaluation of EDP - Entrepreneur development training.

Contact hours required 9

UNIT 4

Production system- Design of production system- Types of production system- Production, planning & control (PPC) - Steps of PPC.

Contact hours required 9

UNIT 5

Communication- Importance of communication system - barriers to communication - listening to people- the power of talk - personal selling - risk taking & resilience - negotiation.

Contact hours required 9

Text Books:

1. Entrepreneurship Development- Sangeeta Sharma, Kindle edition
2. Production & operations Management- Kanishka Bedi,
3. Marketing Management- Philip Kotler.

4. The Business Model Book: Design, build and adapt business ideas that drive business growth: Adam Bock, Gerard George

CoureOutcomes: Attheend of this course students will demonstrate the ability to:

1. Enhancecreativeknowledgeofstudentsregardingselectionofabusinessideaanditsimplem entationprocess.
2. Acquireknowledgeonentrepreneurshipdevelopment,itsPro'sandcon's.
3. AcquirebasicknowledgeonhowtobecomeanEntrepreneur.
4. DevelopknowledgeonProductionsystemsand itssustainabilitythroughproduction,planningandcontrol(PPC)

Developappropriatebusinessmodelandapplyinabetterway.

KHT651 TEXTILE DESIGN AND COLOR LABORATORY

(L-T-P 0-0-2),Credit =1

LIST OF EXPERIMENTS

1) Drawing of lines, horizontal lines, curved lines, lines of growth and grace, thick and thin lines, double lines, broken lines, oblique lines or lines of movement, accented lines.

2) Free hand drawing practice - creation of foliage drawing, small plants and flowers,creepers, birds, animals, clouds and still water.

3) Light and Pigment theory of colours, properties of colour vision, complementary colours, chromatic circle, attributes of colour, properties of colour – hue, value and intensity, simultaneous contrast – contrast of tone and colour harmony.

4) Modification of colours – hue, tone, tint, shade, coloured greys and triadic colour.

5) Drawing of special illusion colour contrasts – mono chromatic contrast and polychromatic contrast.

6) Free hand drawing of elements of textile design – motif, unit figure and design repeat.

7) Creation of textile designs on various bases - rectangular, diamond, ogee, diagonal bases;distribution of unit figures – principles of simple drop and drop reverse; sateen distribution,

all over effects etc.

8) Creation of layout for saree, dhoti, lungi and shawl.

9) Creation of layout for bed sheet, table cover and handloom durries.

10) Documentation of traditional Indian handloom sarees. (Pictures/ samples)

KHT652 FABRIC STRUCTURE LABORATORY

(L-T-P 0-0-2), Credit =1

LIST OF EXPERIMENTS

1) To develop the graph and fabric dealt in fabric structure structure-II.

2) To analyze the compound fabrics – double cloth, Bedford cord, welt, pique, terry, extra warp, extra weft, and leno.

3) To extract fundamental details like count of warp and weft, ends and picks per unit space, warp and weft crimp and weave repeat.

4) To determine the depth of shed and heald shaft movements in tappet shedding mechanism.

5) To prepare the pattern card for dobby shedding mechanism and way in which adjust the depth of shed

6) To study about jacquard shedding mechanism.

7) To sketch different parts of SLSC, DLSC, and DLDC Jacquards and familiarization of their function.

8) To sketch layout of a jacquard loom, Familiarization of mounting of jacquard on a loom.

9) To practice the Harnessing building for straight tie, pointed tie, sectional tie, body border tie.

10) To perform punching, lacing and weaving the design from the prepared graph.

11) To study aboutwarp stop, weft replenishing and positive let-off motion of automatic power loom.

12) To determinethe pick space through 5 and 7 wheel take-up mechanisms.

13) To study about warp protector mechanism, weft detecting and drop box motion of power loom.

14) To perform dismantling, erecting and time setting shedding mechanism, picking (over-picking & under picking) and beat-up mechanism.

15) To prepare the design suitable for saree borders and dhoti borders using Dobbies up to 40 leavers and pegging of lattices as per mounting of Dobbies on power loom.

16) To prepare the album of the samples developed and writing their quality particulars.

KHT 653 FABRIC QUALITY EVALUATION LABORATORY

(L-T-P 0-0-2),Credit =1

LIST OF EXPERIMENTS

Determination of the following:-

1) To determine Fabric tensile strength by using strip strength method.
2) To determine the bursting strength of fabric by using Hydraulic bursting strength tester.
3) To determine the Fabric tear strength by using a (Elmendorf tear) pendulum-type tester.
4) To determine Fabric bending length, flexural rigidity and bending modulus by using stiffness tester.
5) To determine fabric Drape ability by using Cusick drape tester.
6) Measure crease recovery of fabric by crease recovery tester.
7) To determine the fabric thickness by using thickness gauge.
8) Find out the relative abrasion properties of fabrics by Martindale abrasion tester.
9) Fabric pilling resistance Test of pilling of fabrics by ICI Pilling Box Tester.
10) Find out air permeability of fabric by air permeability tester.
11) To determine the fabric GSM by using GSM cutter.